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MAGDEBURG

Peak Detection using Transformers

Autonomous Multisensor Systems Group
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Autonomous Multisensor Systems

Ray Tune Methodology

Define Model, .and Search Space → Training Function → Scheduler Algorithm → Run Ray Tune → Select Best Model

Define Model, .and Search Space

Choose mode LSTM

window_size = [30, 60, 100]

hidden_dim = [8, 16, 32]

split_ratio = [0.2, 0.6, 0.7]

Training Function

MinMaxScaler

Sliding Window

Train model

Eval : Validation metrics each epoch

Reports metrics to Ray Tune

Scheduler Algorithm

27 combinations

Grid Search

ASHAScheduler (Asynchronous Successive Halving Algorithm)

Run trials,
Monitor Performance,
Early-stopping.

Run Ray Tune

Parallel Trial Execution

Resource-efficient search (Scheduler + search algorithm work together)

Save Checkpoints & Results (model weights, metrics)

Select Best Model

Extract best config (hyperparameters)

Extract best result (metrics: R^2, MAE, RMSE)

Ray Tune Report

Current best trial: 7528a_00008 with r2=0.9670910835266113 and params={'window_size': 30, 'hidden_dim': 32, 'split_ratio': 0.7}

Trial name	status	window_size	hidden_dim	split_ratio	iter	total time (s)	r2	mae	rmse
train_lstm_7528a_00000	TERMINATED	30	8	0.2	13	49.733	-0.00374508	0.0322889	0.106449
train_lstm_7528a_00001	TERMINATED	30	16	0.2	4	21.4414	-1.40992	0.111595	0.164943
train_lstm_7528a_00002	TERMINATED	30	32	0.2	1	10.2278	-1.3018	0.114151	0.1612
train_lstm_7528a_00003	TERMINATED	30	8	0.6	21	104.561	0.732741	0.0159571	0.0277545
train_lstm_7528a_00004	TERMINATED	30	16	0.6	16	97.1543	0.482319	0.0181915	0.0386277
train_lstm_7528a_00005	TERMINATED	30	32	0.6	1	13.6324	-0.039005	0.0199964	0.0547238
train_lstm_7528a_00006	TERMINATED	30	8	0.7	4	27.856	0.179096	0.0193439	0.0489695
train_lstm_7528a_00007	TERMINATED	30	16	0.7	18	120.89	0.83067	0.0124545	0.0222406
train_lstm_7528a_00008	TERMINATED	30	32	0.7	40	411.765	0.967091	0.00540482	0.00980474
train_lstm_7528a_00009	TERMINATED	60	8	0.2	1	10.001	-1.56982	0.133539	0.170395
train_lstm_7528a_00010	TERMINATED	60	16	0.2	1	11.6446	-0.597694	0.0821708	0.134355
train_lstm_7528a_00011	TERMINATED	60	32	0.2	1	14.6424	-1.25707	0.111686	0.159691
train_lstm_7528a_00012	TERMINATED	60	8	0.6	1	9.81545	-0.0179611	0.0165596	0.0542136
train_lstm_7528a_00013	TERMINATED	60	16	0.6	4	38.4163	0.25708	0.0200537	0.0463141
train_lstm_7528a_00014	TERMINATED	60	32	0.6	1	17.937	-0.0548892	0.0200867	0.0551881
train_lstm_7528a_00015	TERMINATED	60	8	0.7	1	11.6008	-0.00599051	0.0194208	0.0542742
train_lstm_7528a_00016	TERMINATED	60	16	0.7	4	42.8159	0.316618	0.0185492	0.044733
train_lstm_7528a_00017	TERMINATED	60	32	0.7	1	23.1201	0.00928074	0.0244351	0.0538607
train_lstm_7528a_00018	TERMINATED	100	8	0.2	1	11.0062	-1.66791	0.137362	0.17371
train_lstm_7528a_00019	TERMINATED	100	16	0.2	1	14.2847	0.574573	0.0927346	0.133451

Ray Tune Report Results

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Best config: {'window_size': 30, 'hidden_dim': 32, 'split_ratio': 0.7}

Best result: {'r2': 0.9670910835266113,
'mae': 0.005404820665717125,
'rmse': np.float64(0.009804744152745107),
'training_iteration': 40,
 'date': '2025-10-01_09-43-24',
 'time_total_s': 411.7645902633667}

Posters

https://static.rainfocus.com/nvidia/gtcs25/sess/1734302448222001uamP/poster/P74176-Accelerate%20Cancer%20Research%20With%20AI-Driven%20Multimodal%20Data%20Integration_1742357548522001hXfL.pdf

https://static.rainfocus.com/nvidia/gtcs25/sess/1728896533468001XRsY/poster/P73390-Multi-Location%20Solar%20Irradiance%20Forecasting%20Using%20Hybrid%20Quantum%20Neural%20Networks%20for%20Renewable%20Energy%20Assessment_1742363713617001gGF2.pdf

https://www.nvidia.com/gtc/poster-gallery/?utm_source=chatgpt.com

Transformer Model for Peak Detection

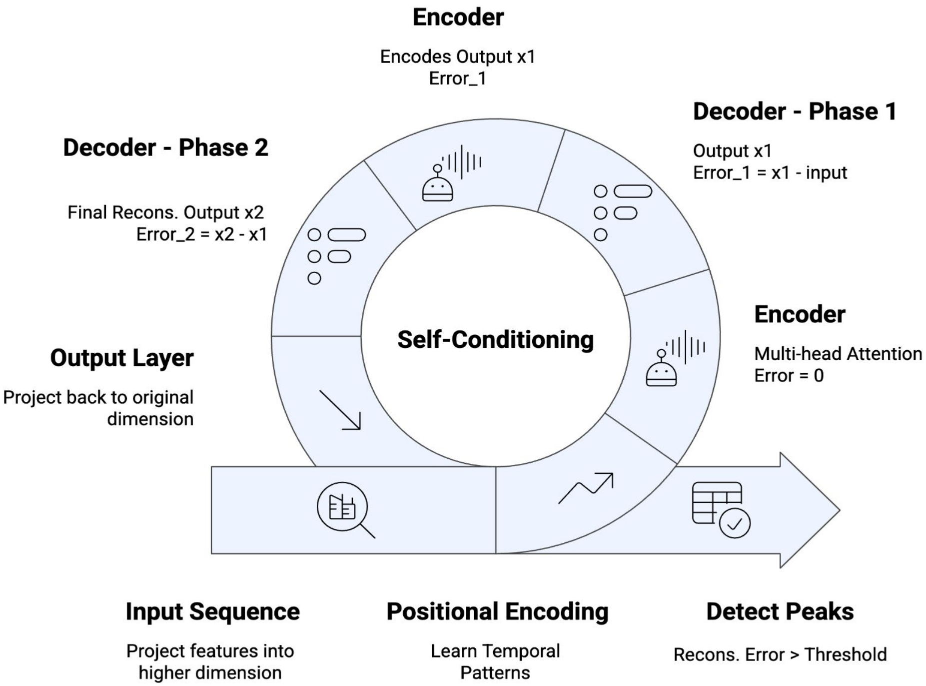


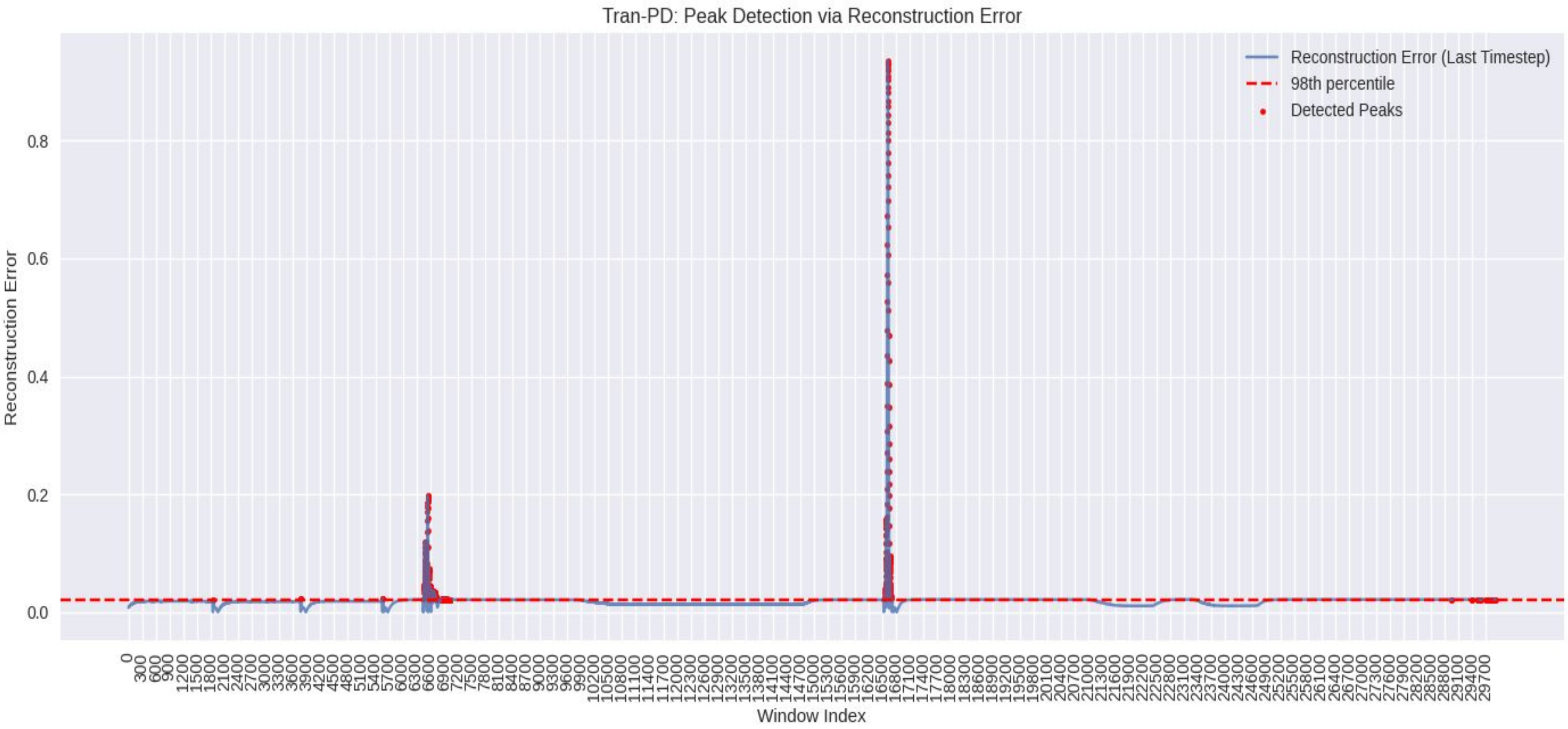
Fig : TranPD Model

- Peak = Identifying local maxima or minima
- 2017 Paper **- Attention Is All You Need!!!**
- Powers models like BERT, GPT, ViT etc.
- NLP, Computer Vision, Time-Series Forecasting

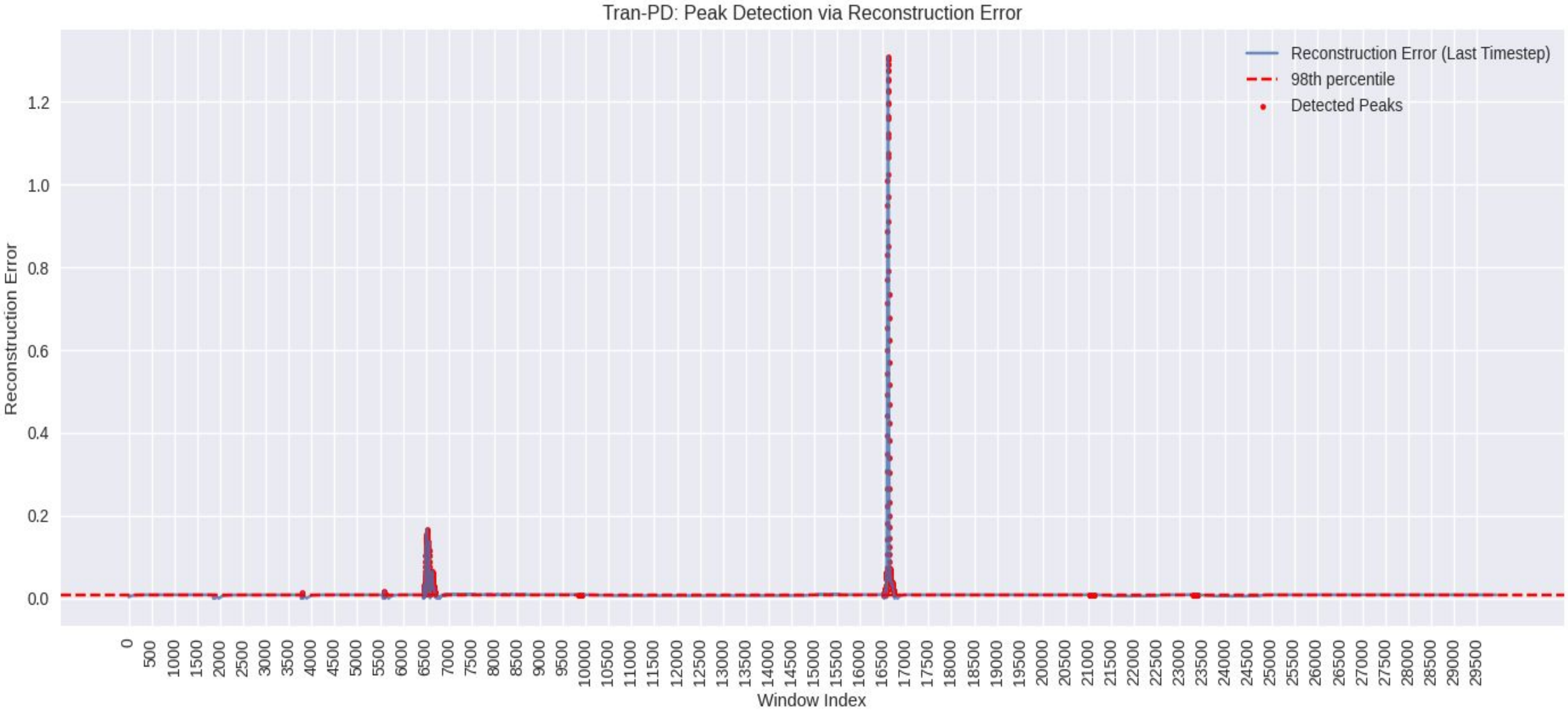
Regression Model Workflow

Train Test Split	Training : Reconstruction	Random Forest with Loss	Random Forest without Loss	MAE, R^2
Chronological Train : Test = 25173: 16783	W_s = 60 Dim = 16 Trained Model with Training Set Reconstruction on Training Set + Reconstruction on Test Set RE for complete Dataset	Num of Features = 53 Target = Current | 6 n_estimators = 100	Num of Features = 52 Target = Current | 6 n_estimators = 100	Comparing Accuracy for Both.

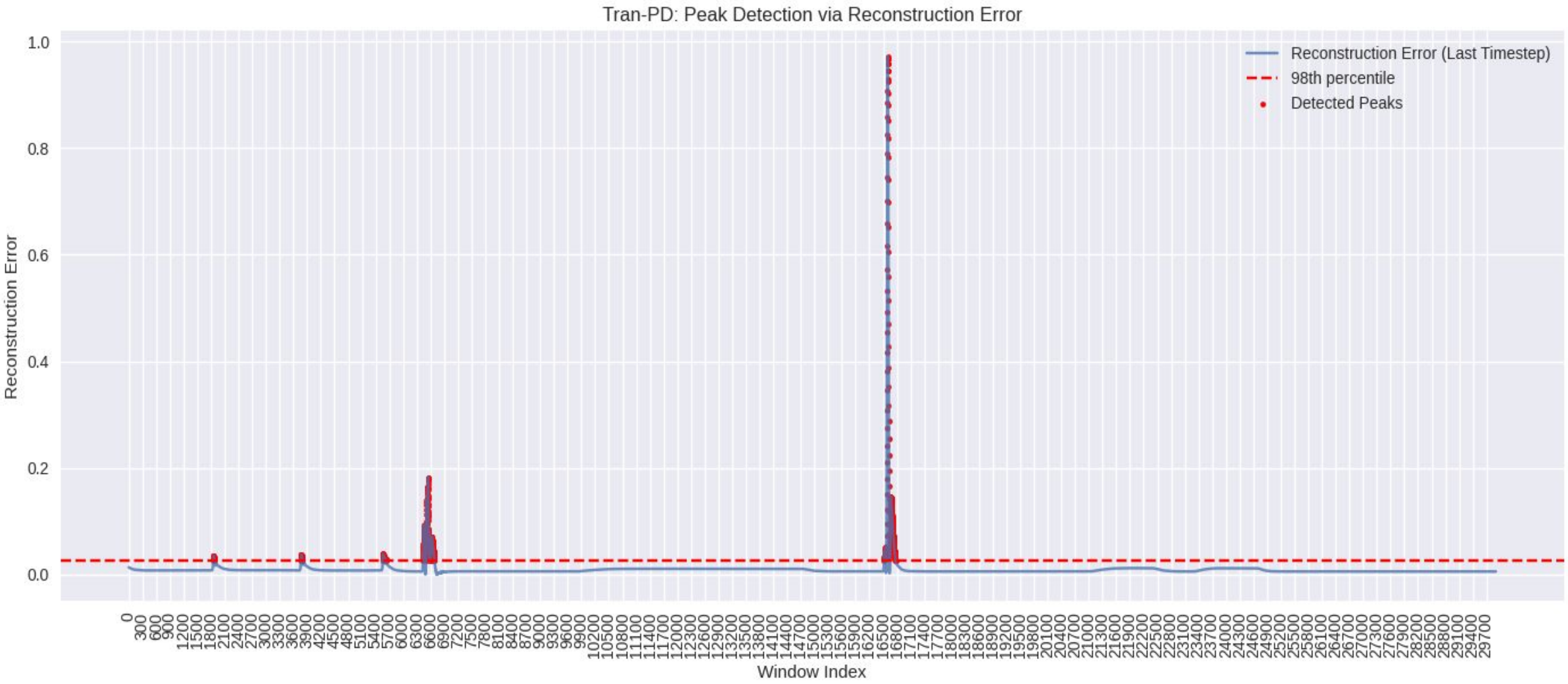
CMX-AL Dataset Results - 587 Peaks, W=30, dim=32



CMX-AL Dataset Results - 599 Peaks, W=30, dim=16



CMX-AL Dataset Results - 599 Peaks, W=60, dim=16



Regression With Loss vs Without Loss

	Num of Features	MAE	RMSE	R^2
RF with Loss	53	0.1128	0.8497	0.9340
RF without Loss	52	0.1244	0.8379	0.9358

Results for Train:Test Split

Dataset, Size	Train:Test Split Ratio	Training Size	Test Size	W_size, Dim	No. of Peaks
DMC2_S_CP2, 42016	60: 40	25192	16794	W_size=30 Dim=16	336
DMC2_S_CP2, 42016	40:60	16794	25192	W_size=30 Dim=16	504
DMC2_S_CP2, 42016	20:80	8396	33588	W_size=30 Dim=16	672
CMX1_AL_CP2, 74822	60:40	44857	29905	W_size=30 Dim=16	599
CMX1_AL_CP2, 74822	60:40	44857	29905	W_size=60 Dim=16	599

Autoencoder Reconstruction Loss

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Train Test Split	Training : AE Reconstruction	Random Forest with AE Loss	Random Forest without AE Loss	MAE, R^2
Chronological Train : Test = 60 : 40 = 25173: 16783	Trained Model with Training Set Reconstruction on Training Set + Reconstruction on Test Set RE for complete Dataset	With ZScore and Tran Loss as well Num of Features = 55 Target = Current | 6 n_estimators = 100	Num of Features = 52 Target = Current | 6 n_estimators = 100	Comparing Accuracy for Both.

Results - Tran Loss, Zscore, AE Loss as a Feature

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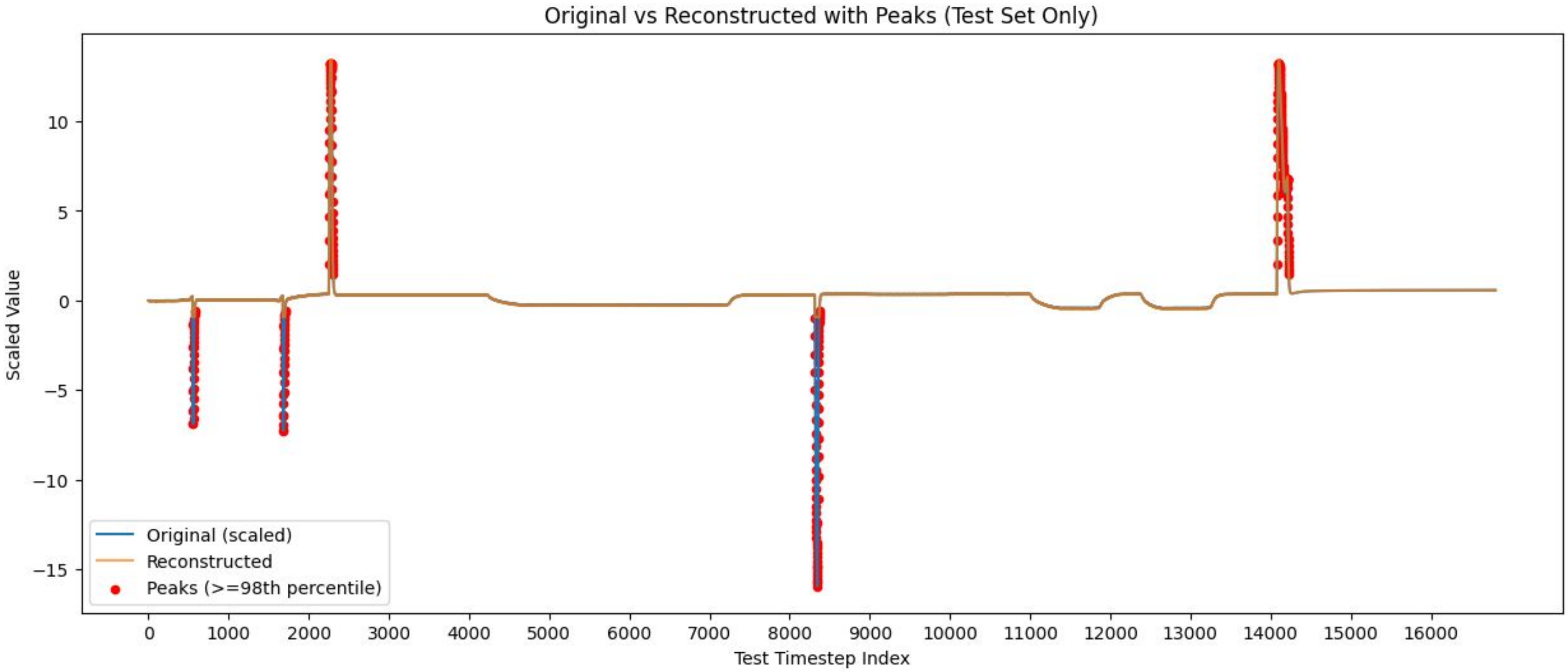
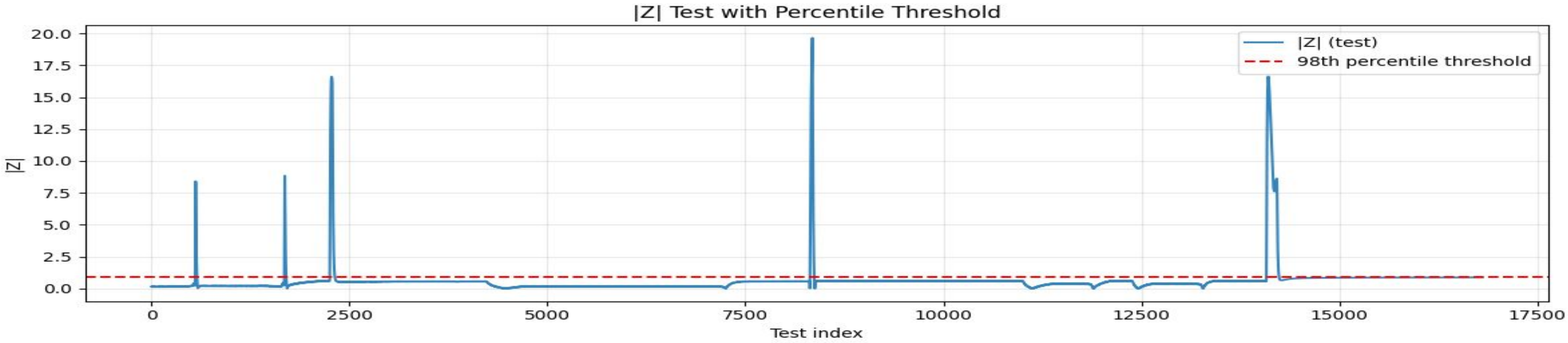
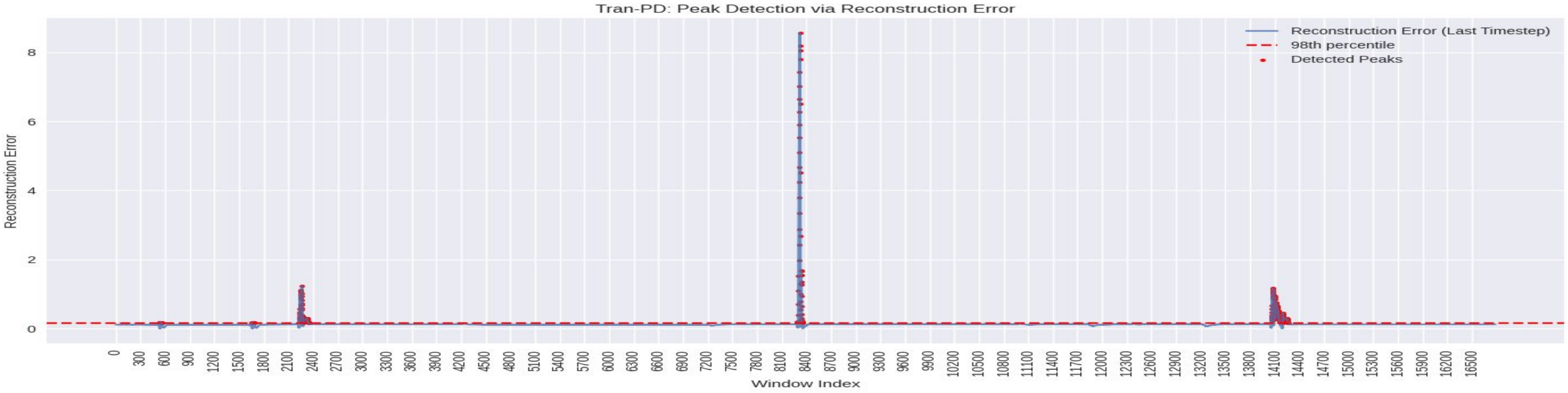
Dataset	Features	MAE	RMSE	R^2
DMC2_S_CP2	52	0.1244	0.7020	0..9358
DMC2_S_CP2	52 + RE	0.1128	0.8497	0.9340
DMC2_S_CP2	52 + Zsore	0.3262	0.5711	0.9702
DMC2_S_CP2	52 + RE + Zsore	0.0465	0.3437	0.9686
DMC2_S_CP2	52 + Tran RE + Zsore + AE RE	**0.0436**	0.5633	**0.9709**

Z-Score vs. Transformer Model



Z-Score vs. Transformer Model





Z-Score vs. Transformer Model

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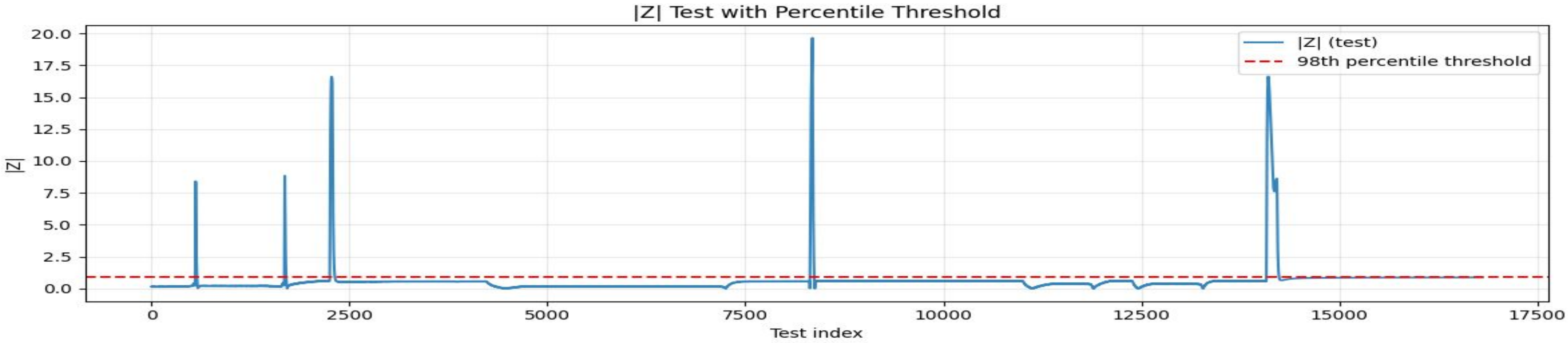
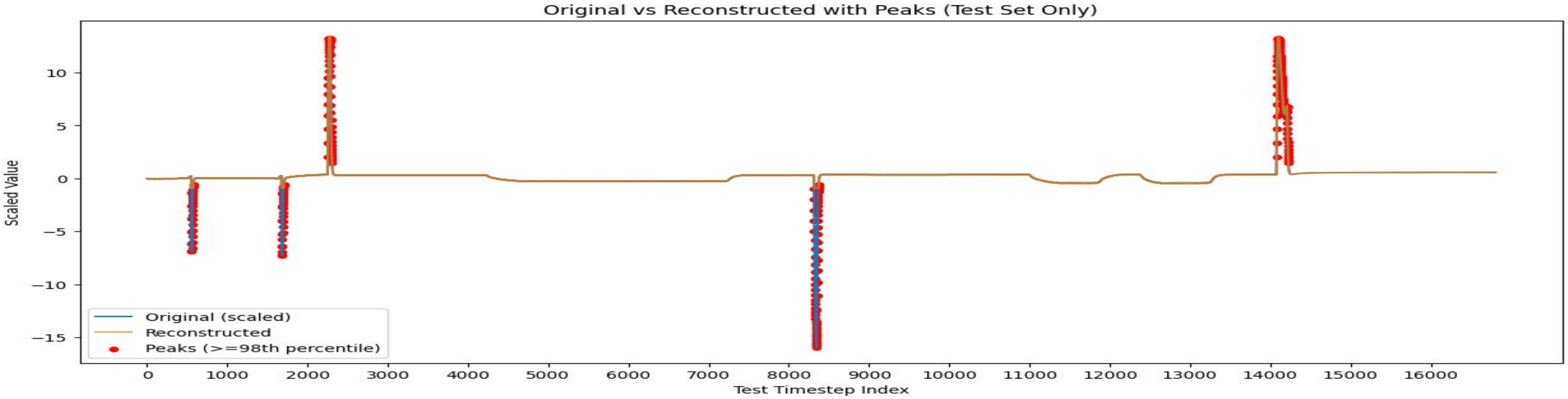
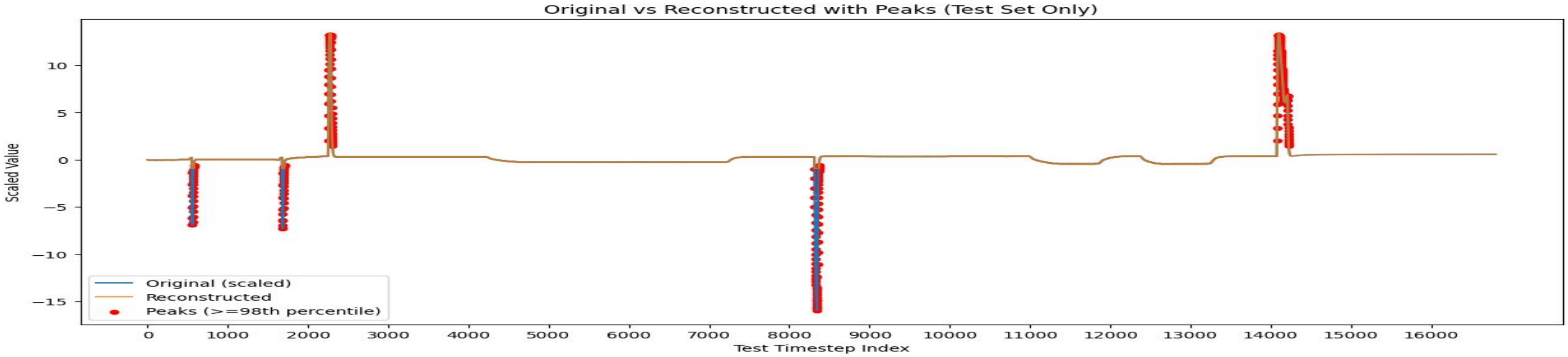
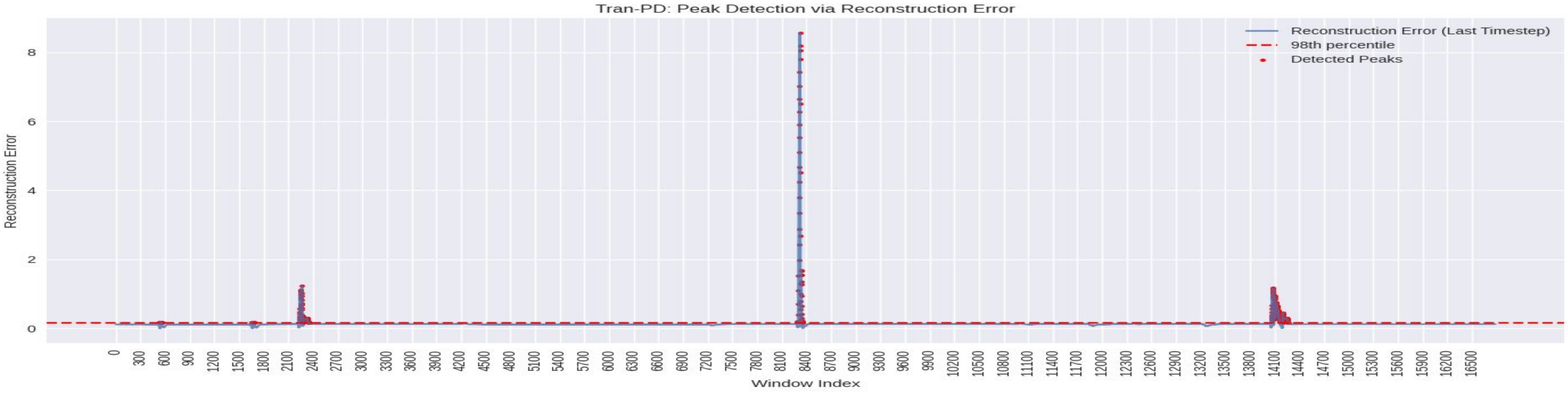




Z-Score vs. Transformer Model





Z-Score vs. Transformer Model

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Todo

1. RMsE R2, of RF AND LSTM with 20 80 te=rain test split - prediction
2. Peak detection
    a. arch AE
    b. AE hyperparams And
    c. AE No of PEaks for each params
    d. arch Zscore
    e. Zscore hyperparams And
    f. Zscore No of PEaks for each params
3. Original Data AE loss and Zscore as feature , 20 80 , RMSE and MAE , R2

AE Architecture

Input (window_size, input_dim)

Encoder: 1 Dense layer (ReLU)

Decoder: 1 Dense layer (Linear)

AE Hyperparams

Hyperparameter	Values
Latent Dim	[8, 16, 32, 64]
Win Size ; **Step (stride)**	[30, 60, 100], Step : [5, 10]
Batch size	32
Epochs, Optimizer	50 , Adam , Learning rate = 0.001
Loss Function	MSE
Activation (Encoder)	ReLu
Activation (Encoder)	Linear
Scaler	MinMaxscaler
Threshold	95th Percentile of reconstruction errors

AE No of Peaks for each param

Information saved in dataset with Column Names : Dataset, Window_Size , Step Latent_Space, Num_Peaks

This is saved in a dataset here:
https://github.com/shweta27407/TranPD/blob/main/Meenu_Autoencoder/Meenu_Autoencoder_tuned_Peaks_without_peak_indices.csv

Z-Score vs. Transformer Model

Todo

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   a. arch AE
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Thank You For Your Attention!

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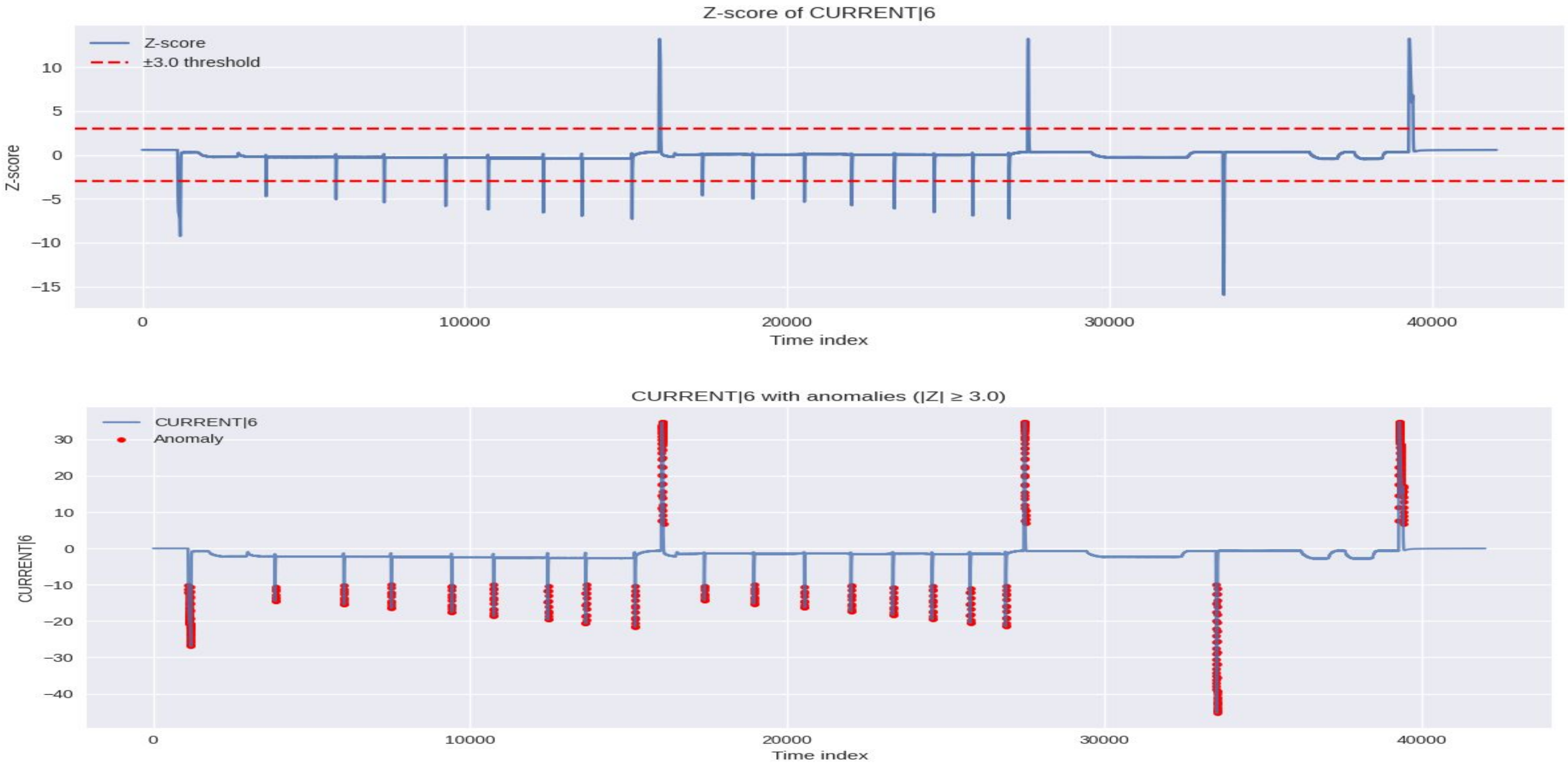
Result Analysis - Statistical Summary

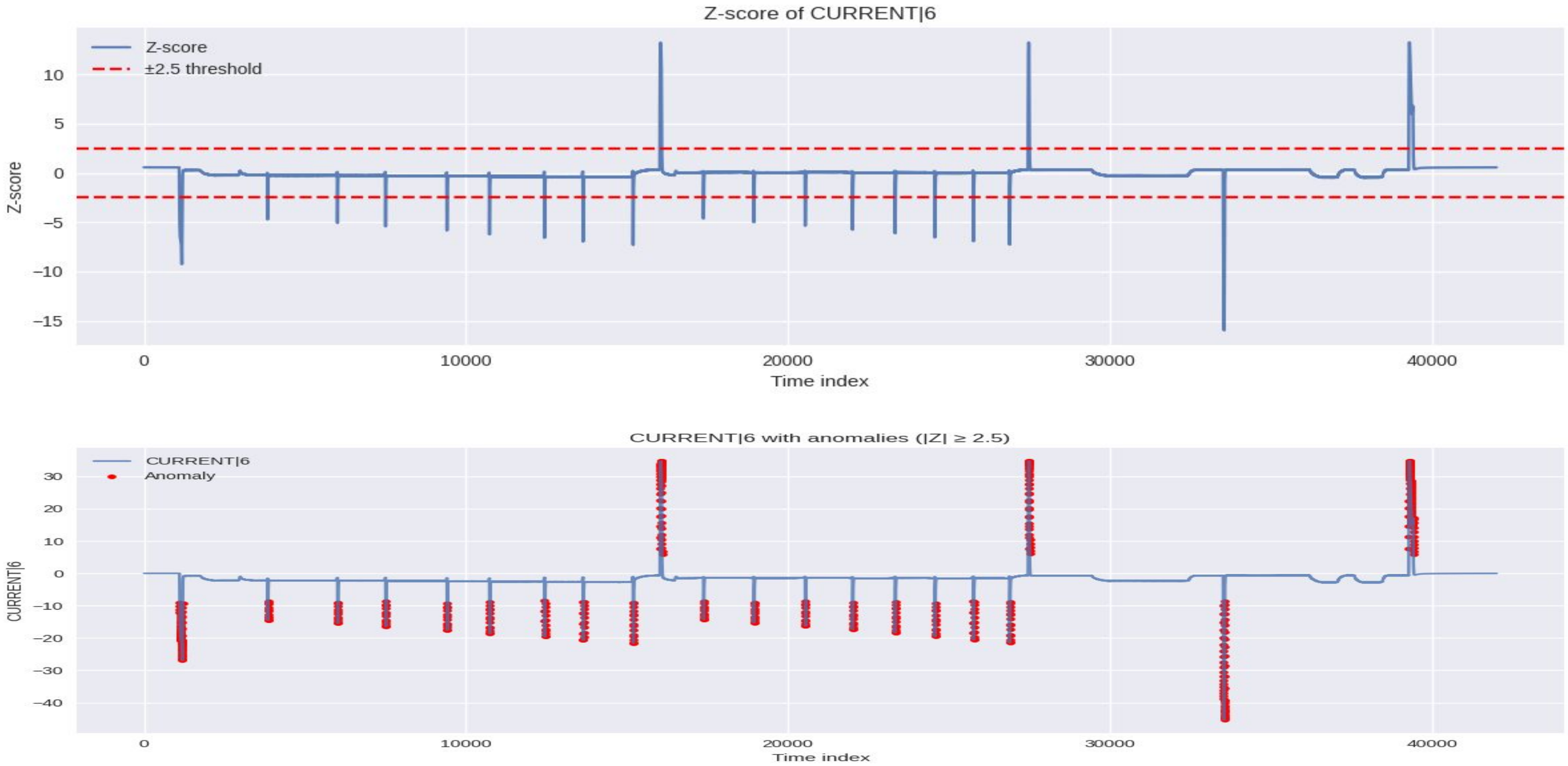
Measure	W = 120, D= 64	W = 120, D= 32	W = 120, D= 16	W = 60, D= 32	W = 60, D= 16	W = 30, D= 32	W = 30, D= 16
No. of Peaks	336	336	336	336	336	336	336
Threshold	0.82	0.11	0.08	0.23	0.15	0.17	0.06
Mean RE	0.12	0.03	0.07	0.12	0.13	0.03	0.02
Min RE	0	0.00	0	0	0	0	0 - perfect zero
Max RE	10.44	7.50	10.55	8.558	8.559	9.33	9.40

Table : Statistical Analysis of Reconstruction Error

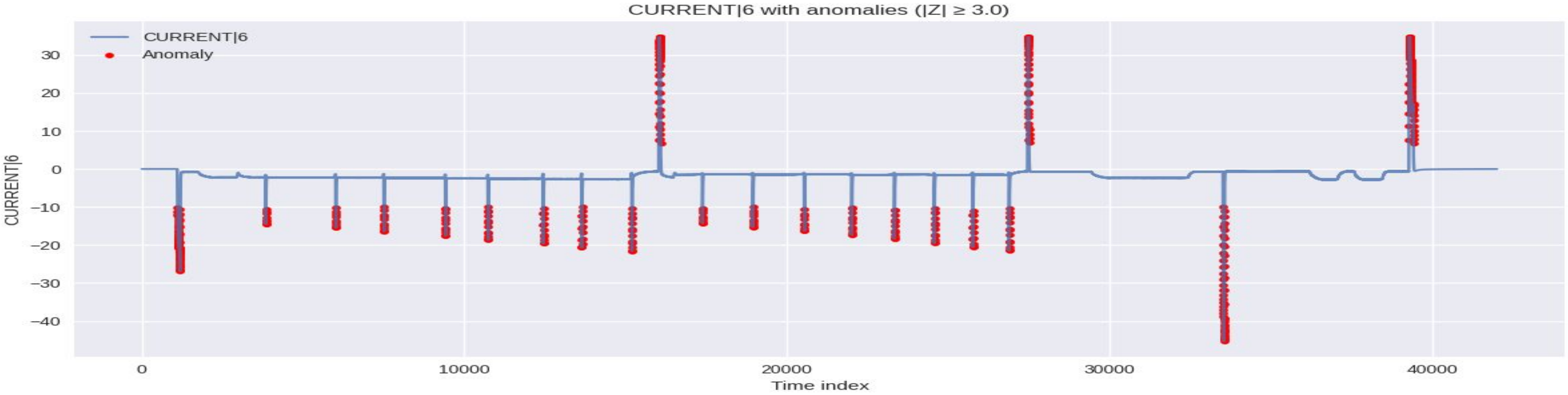
Z-Score : Threshold - 3, Num of Peaks = 548

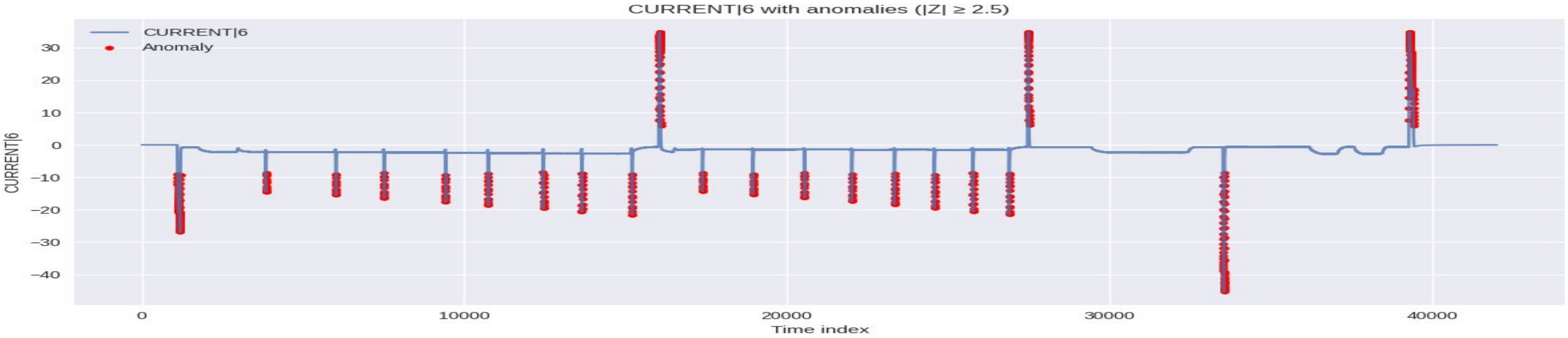


Z-Score : Threshold - 2.5, Num of Peaks = 586



Z-Score : Threshold - 3 Vs. Threshold - 2.5





Z-Score vs. Transformer Model

	Z-Score	TranformerPD
No of Peaks - Test Dataset	396	336
Mean	Z-Score : 0.6360	RE Mean - 0.13
Threshold 98%	Z-score Threshold - 0.8816	RE Threshold - 0.1565

Future Work

1. Regression Analysis

2. Implement Adaptive Thresholding Mechanism

3. Multivariate Peak Detection

4. Peak Forecasting

5. regression using Loss and without Loss try to predict current and later on all cure=rent features

6. z score vs DLM - Prediction power (R2)

References

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3. https://github.com/imperial-qore/TranAD
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5. https://stackoverflow.com/questions/22583391/peak-signal-detection-in-realtime-timeseries-data